CP2 Biochemistry: top 50 questions

Lecture 27 · Cholesterol: properties & synthesis (+ Primer 11)

HMG-CoA synthase: ketones vs cholesterol

1. *Heavy 2026 objective*
If you have an HMG-CoA produced in the mitochondria, what do you expect to synthesize?

A. Cholesterol
B. Ketone bodies
C. Mevalonate
D. Glucose

Answer: B. Ketone bodies

HMG-CoA (3-hydroxy-3-methylglutaryl-CoA) made by the **mitochondrial** HMG-CoA synthase goes to **ketone bodies**: HMG-CoA lyase splits it into acetoacetate (→ β-hydroxybutyrate, acetone).
A (cholesterol), C (mevalonate): come from **cytosolic** HMG-CoA, reduced by HMG-CoA reductase on the ER (endoplasmic reticulum).
D (glucose): HMG-CoA is not a glucose precursor.

HMG-CoA reductase and statins

2. *Faculty asked it 3× in 2026*
Many pharmaceutical drugs achieve their effect by inhibiting key enzymes in the human body. An example of a widely used drug that is a competitive inhibitor of the enzyme HMG CoA Reductase is:

A. Steroids
B. Aspirin
C. Orlistat
D. Lovastatin
E. Ibuprofen

Answer: D. Lovastatin

Lovastatin is a statin: a structural look-alike of HMG-CoA (HMG-CoA = 3-hydroxy-3-methylglutaryl-CoA) that **competitively inhibits HMG-CoA reductase**, the rate-limiting enzyme of cholesterol synthesis.
A (steroids): glucocorticoids inhibit phospholipase A2, not HMG-CoA reductase.
B (aspirin), E (ibuprofen): inhibit COX (cyclooxygenase); aspirin irreversibly, ibuprofen reversibly.
C (orlistat): inhibits pancreatic lipase in the gut.

3. *2026 self-assessment*
Statin drugs (e.g., Atorvastatin, Rosuvastatin) act to decrease plasma cholesterol. This class of drugs function by:

A. Inhibiting the synthesis of HMG-CoA
B. Stimulating the clearance of HDL particles
C. Inhibiting the synthesis of mevalonate
D. Binding bile acids in the gut
E. Inhibiting HMG-CoA lyase

Answer: C. Inhibiting the synthesis of mevalonate

Statins competitively inhibit **HMG-CoA reductase**, the enzyme that turns HMG-CoA (HMG-CoA = 3-hydroxy-3-methylglutaryl-CoA) into **mevalonate**. So they block mevalonate synthesis.
A: HMG-CoA is made one step earlier by HMG-CoA synthase, which statins do not touch.
D: binding bile acids in the gut is what bile-acid sequestrants (cholestyramine, colestipol) do.
E: HMG-CoA lyase makes ketone bodies in mitochondria.
B: statins lower LDL (low-density lipoprotein); they do not clear HDL (high-density lipoprotein).

SREBP2 sterol-dependent regulation

4. *Heavy 2026 objective*

Which of the following describes regulation of cholesterol synthesis during low cholesterol states?

- **A.** SREBP2 translocates to the nucleus to increase synthesis of HMG-CoA reductase
- **B.** Glucagon is upregulated to increase cholesterol levels
- **C.** Insulin down regulated to increase cholesterol synthesis
- **D.** SREBP2 remains sequestered in the endoplasmic reticulum membrane
- **E.** Thyroxine is downregulated to increase cholesterol synthesis

Answer: A. SREBP2 translocates to the nucleus to increase synthesis of HMG-CoA reductase

Low cholesterol → SREBP2 (sterol regulatory element-binding protein 2) moves from the ER (endoplasmic reticulum) to the Golgi, is cut, and its fragment enters the **nucleus** → ↑ HMG-CoA reductase synthesis.
D: staying in the ER membrane is the **high**-cholesterol state.
B: glucagon ↓ HMG-CoA reductase expression, so it lowers synthesis.
C, E: insulin and thyroxine ↑ the enzyme; less of them would lower synthesis. Hormonal control is independent of the sterol level.

Bile acid synthesis, 7α-hydroxylase, enterohepatic circulation

5. *Faculty asked it 2× in 2026*

A 39-year-old obese woman presenting with severe abdominal pain was diagnosed with cholelithiasis and underwent an elective cholecystectomy. Decreased activity of which of the following enzymes is the most likely cause of her cholelithiasis?

- **A.** HMG-CoA reductase
- **B.** 7α-Hydroxylase
- **C.** HMG-CoA lyase
- **D.** HMG-CoA synthase
- **E.** Thiolase

Answer: B. 7α-Hydroxylase

7α-Hydroxylase (CYP7A1) is the rate-limiting enzyme that turns **cholesterol into bile acids**. Low activity → too few bile salts for the cholesterol in bile → bile **supersaturated with cholesterol** → crystals → cholesterol gallstones.
A (HMG-CoA reductase): low activity means **less** cholesterol made, which would protect against stones.
C (HMG-CoA lyase): makes ketone bodies.
D (HMG-CoA synthase): makes HMG-CoA; low activity lowers cholesterol.
E (thiolase): joins two acetyl-CoA; not linked to stones.
HMG-CoA = 3-hydroxy-3-methylglutaryl-CoA.

6. *2026 self-assessment*

A 45-year old male presents for a routine physical. His lipid profile shows a serum total cholesterol of 500 mg/dL and LDL cholesterol of 350 mg/dL. Physician prescribes atorvastatin and colestipol. How does colestipol contribute to the cholesterol lowering effect?

- **A.** By inhibiting bile acid synthesis
- **B.** By blocking the recycling of bile acids to liver
- **C.** By inhibiting HMG-CoA reductase activity
- **D.** By preventing rhabdomyolysis
- **E.** By preventing the excretion of cholesterol

Answer: B. By blocking the recycling of bile acids to liver

Colestipol is a **bile-acid sequestrant**: a resin that binds bile acids in the gut so they leave in stool instead of returning to the liver (it blocks the **enterohepatic circulation**). The liver then turns more cholesterol into bile acids (↑ 7α-hydroxylase), liver cholesterol falls, LDL (low-density lipoprotein) receptors rise, blood LDL falls.
A: the opposite; bile acid synthesis ↑.
C: that is atorvastatin's action.
E: it ↑ cholesterol excretion (as bile acids).
D: it does not prevent statin muscle injury.

7. *2026 Canvas practice quiz*

A 69-year old woman with high LDL cholesterol was prescribed a bile acid sequestrant (cholestyramine) as part of her treatment regimen to lower blood cholesterol levels. This treatment most likely results in increased flux through what metabolic pathway in the liver?

A. Conversion of cholesterol to bile acids

B. Fatty acid and triacylglycerol synthesis

C. Fatty acid desaturation

D. Gluconeogenesis

E. Production and secretion of VLDL particles

Answer: A. Conversion of cholesterol to bile acids

Cholestyramine binds bile acids in the gut, so fewer return to the liver. Less bile acid means less feedback through FXR (farnesoid X receptor), so **7α-hydroxylase rises** and the liver **converts more cholesterol into bile acids**. Liver cholesterol falls → ↑ LDL (low-density lipoprotein) receptors → ↓ blood LDL.
E: VLDL (very-low-density lipoprotein) output can rise a little (triglycerides may go up), but the direct, large change is bile acid synthesis.
B, C, D: fatty acid synthesis, desaturation and gluconeogenesis are not driven by bile acid loss.

8. *2026 Canvas practice quiz*

40 y/o woman is presented with discomfort in right upper quadrant after eating. Ultrasound shows small cholesterol gallstones. She begins oral therapy with ursodeoxycholic acid. What is the goal of this treatment?

A. To add bilirubin to bile

B. To decreased concentration of bile acids

C. To decreased ratio of cholesterol to bile acids

D. To increased in the ability of the gallbladder to concentrate bile

E. To increased in the concentration of lecithin in bile

Answer: C. To decreased ratio of cholesterol to bile acids

Ursodeoxycholic acid ↓ cholesterol secretion into bile and adds to the bile acid pool, so it **lowers the cholesterol-to-bile-acid ratio**. Bile can then hold cholesterol in micelles, and small cholesterol stones slowly dissolve.
B: it raises, not lowers, bile acids relative to cholesterol.
A: bilirubin is a pigment; adding it would favor pigment stones.
D: more concentrated bile would favor stones.
E: its effect is on cholesterol and bile acids, not on lecithin (phosphatidylcholine).

Lecture 28 · Phospholipids & sphingolipids: medical aspects

Lung surfactant and respiratory distress

9. *Faculty asked it 3× in 2026*

Carol is a 28-year-old pregnant female at 30 weeks of gestation admitted for monitoring of progress of labor. Based on a test performed using amniotic fluid, the obstetrician decided to administer corticosteroids to the mother to avoid the risk of respiratory distress in the premature baby. Which of the following test results would have prompted the doctor to make that decision?

A. Sphingomyelin : DPPC < 2.0

B. Lecithin : Phosphatidylserine > 2.0

C. Lecithin : DPPC < 2.0

D. Lecithin : Sphingomyelin < 2.0

E. Lecithin : Phosphatidylserine < 2.0

Answer: D. Lecithin : Sphingomyelin < 2.0

An amniotic **lecithin : sphingomyelin (L:S) ratio <2** means the fetal lungs are **immature** (too little surfactant), so the mother gets corticosteroids to speed fetal surfactant synthesis.
A, C (ratios with DPPC): DPPC (dipalmitoyl phosphatidylcholine) is itself the main lecithin; the test compares lecithin with sphingomyelin.
B, E (lecithin : phosphatidylserine): not the clinical test; sphingomyelin is the reference lipid because its level stays steady.

10. *2026 self-assessment*

Carol is a 28-year old pregnant female at 30 weeks of gestation admitted for monitoring of progress of labor. Based on a test performed using amniotic fluid, the obstetrician decided to administer dexamethasone to the mother to avoid the risk of respiratory distress in the premature baby. Dexamethasone induces the synthesis of which of the following compounds?

A. Phosphatidylethanolamine

B. Sphingomyelin

C. Cardiolipin

D. Phosphatidylserine

E. DPPC

Answer: E. DPPC

Dexamethasone (a glucocorticoid given to the mother) speeds maturation of fetal **type II alveolar cells** and induces synthesis of **DPPC** (dipalmitoyl phosphatidylcholine), the main surfactant lipid.
B (sphingomyelin): the steady reference lipid in the lecithin : sphingomyelin ratio; steroids do not target it.
A, D (phosphatidylethanolamine, phosphatidylserine): ordinary membrane phospholipids, not surfactant.
C (cardiolipin): inner mitochondrial membrane lipid.

Glycosphingolipid functions and classes

11. *2026 self-assessment*
Which of the following is a glycosphingolipid?

A. Sphingomyelin
B. Lysolecithin
C. Lecithin
D. Cerebroside
E. Phosphatidylcholine

Answer: D. Cerebroside

A **cerebroside** = ceramide + 1–2 sugars (glucose or galactose), so it is a **glycosphingolipid**.
A (sphingomyelin): ceramide + phosphocholine; a sphingo*phospho*lipid with no sugar.
B (lysolecithin): phosphatidylcholine missing one fatty acid (made by phospholipase A2); glycerol-based.
C, E (lecithin = phosphatidylcholine): glycerophospholipid; no sphingosine, no sugar.

12. *2026 Canvas practice quiz*
Which of the following structures is common to all sphingolipids?

A. Carnitine
B. Ceramide
C. Diacylglycerol
D. Sphingomyelin
E. Choline

Answer: B. Ceramide

Ceramide (sphingosine + an amide-linked fatty acid) is the core of every sphingolipid: sphingomyelin adds phosphocholine; glycosphingolipids add sugars.
D (sphingomyelin): one sphingolipid itself, not the shared part.
E (choline): only in sphingomyelin (and phosphatidylcholine).
C (diacylglycerol): backbone of glycerophospholipids, not sphingolipids.
A (carnitine): carries fatty acids into mitochondria.

Phospholipid biosynthesis pathways

13. *2026 Canvas practice quiz*
Which of the following phospholipids can be converted into phosphatidylcholine by an S-adenosylmethionine- dependent enzyme?

A. Sphingomyelin
B. Phosphatidylserine
C. Phosphatidylethanolamine
D. Phosphatidylinositol
E. Lecithin

Answer: C. Phosphatidylethanolamine

Phosphatidylethanolamine (PE) gets three methyl groups from **S-adenosylmethionine (SAM)** (PE N-methyltransferase) → phosphatidylcholine (PC); the liver uses this when choline is low.
B (phosphatidylserine): decarboxylated to PE, not methylated to PC.
E (lecithin): is PC itself, the product.
A (sphingomyelin): a sphingolipid; it takes its phosphocholine from PC.
D (phosphatidylinositol): no amine to methylate.

Sphingomyelin and its degradation

14. *2026 Canvas practice quiz*
The backbone of ceramide is derived from a long-chain fatty-acyl CoA and

A. Serine
B. Glycine
C. Diacylglycerol
D. Phosphatidic acid
E. Dihydroxyacetone phosphate

Answer: A. Serine

Ceramide synthesis starts with **palmitoyl-CoA** (CoA = coenzyme A) **+ serine** → sphinganine; a second fatty acyl-CoA is then joined by an amide bond → ceramide.
B (glycine): glycine + succinyl-CoA starts heme synthesis, not sphingolipids.
C/D (diacylglycerol, phosphatidic acid): the glycerol backbone of glycerophospholipids such as lecithin, not of sphingolipids.
E (dihydroxyacetone phosphate): source of the glycerol backbone of plasmalogens and triacylglycerol.

Niemann-Pick disease

15. *2026 self-assessment*

A 15-year old female was brought to the clinic for fever and productive cough. Past history reveals recurrent lung infections, easy bruising, and being easily fatigued while playing. On PE, she has hepatosplenomegaly. Result of the fundoscopy is shown below (figure A). Liver biopsy of Kupffer cells showed foamy inclusions in the lysosomes (figure B). What is her most likely diagnosis?

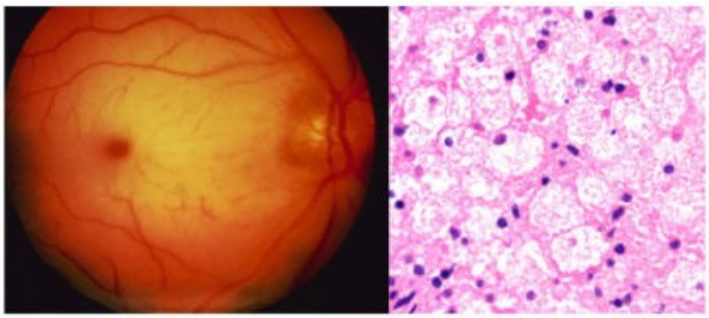


figure A figure B

A. Krabbe disease

B. Fabry disease

C. Gaucher disease

D. Tay-Sachs disease

E. Niemann-Pick disease

Answer: E. Niemann-Pick disease

Figure A: **cherry-red spot on the macula**. Figure B: **foamy, lipid-filled macrophages** (Kupffer cells). Add **hepatosplenomegaly** = **Niemann-Pick** (sphingomyelin storage). Reaching age 15 with lung infections fits milder **type B**.

C (Gaucher): hepatosplenomegaly and bruising too, but no cherry-red spot; its macrophages look like crumpled tissue paper, not foamy.

D (Tay-Sachs): cherry-red spot, but normal liver and spleen; death by age 3.

A (Krabbe), B (Fabry): neither has a cherry-red spot or hepatosplenomegaly.

Tay-Sachs disease

16. *2026 self-assessment*
Post-mortem lipid analysis of the brain from a 2-year old male child demonstrated the accumulation of GM2-gangliosides. The clinical history stated that the child's parents were first cousins, and that he had deteriorated physically and mentally over the past year, and lost his vision, was unable to eat by himself, and had a cherry red spot on the macula. A deficiency in which of the following enzymes is likely to cause his pathological condition?

A. α-Galactosidase A
B. Hexosaminidase A
C. Sphingomyelinase
D. β-Glucocerebrosidase
E. β-Galactocerebrosidase

Answer: B. Hexosaminidase A

GM2 ganglioside in the brain + cherry-red macula + loss of vision and skills (parents are cousins = autosomal recessive) = **Tay-Sachs** → **hexosaminidase A** deficiency; this enzyme removes N-acetylgalactosamine from GM2.
C (sphingomyelinase): Niemann-Pick; stores sphingomyelin.
A (α-galactosidase A): Fabry; stores globotriaosylceramide.
D (β-glucocerebrosidase): Gaucher; stores glucocerebroside.
E (β-galactocerebrosidase): Krabbe; stores galactocerebroside.

17. *Faculty asked it 3× in 2026*
A 2-year-old child who is in the 50th percentile for height and weight, is unable to sit by himself or talk and has an exaggerated startle response. Fundoscopic examination shows a cherry red spot on the macula. Liver ultrasound is normal. Which of the following lipids is likely accumulating in his neural tissue?

A. Sulfatides
B. Cholesterol
C. Sphingomyelin
D. Glucocerebroside
E. Ganglioside GM2

Answer: E. Ganglioside GM2

Cannot sit or talk, **exaggerated startle**, **cherry-red macula** and a **normal liver** = **Tay-Sachs** (hexosaminidase A deficiency) → **GM2 ganglioside** accumulates in neurons.
C (sphingomyelin): Niemann-Pick; cherry-red spot too, but with hepatosplenomegaly.
D (glucocerebroside): Gaucher; hepatosplenomegaly and bone disease.
A (sulfatides): metachromatic leukodystrophy; demyelination, no cherry-red spot.
B (cholesterol): trapped in Niemann-Pick type C, which enlarges the liver and spleen.

Gaucher disease

18. *Faculty asked it 2× in 2026*
A 7-year-old boy presents with fatigue, easy bruising, bone pain, and hepatosplenomegaly. Laboratory studies reveal anemia and thrombocytopenia. Bone marrow biopsy shows macrophages with a characteristic “crumpled tissue paper” appearance. Deficiency of which of the following enzymes is most likely responsible for this condition?

A. Acid sphingomyelinase
B. Hexosaminidase A
C. α-Galactosidase A
D. β-Glucocerebrosidase
E. Arylsulfatase A

Answer: D. β-Glucocerebrosidase

Bone pain, **hepatosplenomegaly**, anemia and low platelets (marrow crowded out) with **'crumpled tissue paper' marrow macrophages** = **Gaucher disease** → **β-glucocerebrosidase** deficiency (glucocerebroside storage).
A (acid sphingomyelinase): Niemann-Pick; foamy macrophages, often a cherry-red macula.
B (hexosaminidase A): Tay-Sachs; brain disease, no organ enlargement.
C (α-galactosidase A): Fabry; angiokeratomas, painful extremities, kidney failure.
E (arylsulfatase A): metachromatic leukodystrophy; demyelination.

Fabry disease

19. *2026 Canvas practice quiz*
An adult male suffered from progressive heart failure, hearing loss and repeated thromboembolism. Upon his death at age 63, autopsy revealed cardiomyopathy and elevated cardiac storage of globotriaosylceramide. Glycolipid storage in other tissues was not observed. Angiokeratomas were noted on the chest. The lipid storage disease in this patient is most likely to be:

A. Fabry disease

B. Gaucher disease

C. Niemann-Pick B

D. Tay-Sachs disease

E. Niemann-Pick A

Answer: A. Fabry disease

Heart failure (cardiomyopathy), hearing loss, **angiokeratomas** and stored **globotriaosylceramide** (= ceramide trihexoside) = **Fabry disease** (α-galactosidase A deficiency). Later-onset forms can hit mainly the heart.
B (Gaucher): glucocerebroside in macrophages; hepatosplenomegaly, bone disease.
C/E (Niemann-Pick B/A): sphingomyelin; hepatosplenomegaly (type A also brain disease, death by age 2–3).
D (Tay-Sachs): GM2 ganglioside in neurons; death in early childhood.

Lecture 29 · Lipoproteins (+ Primer 12)

Key enzymes: CETP, LCAT, ACAT

20. *2026 Canvas practice quiz*
Which of the following functions is the Lecithin cholesterol acyltransferase (LCAT) enzyme involved in?

A. Converting cholesterol to cholesterol esters

B. Transfer of cholesterol to the other lipoproteins

C. Degradation of HDL in liver

D. Degradation of chylomicrons

E. Decreased uptake of cholesterol by hepatocytes

Answer: A. Converting cholesterol to cholesterol esters

LCAT (lecithin–cholesterol acyltransferase), activated by apo A-I, moves a fatty acid from lecithin (phosphatidylcholine) onto cholesterol → **cholesteryl ester**. This traps cholesterol in the core of HDL (high-density lipoprotein).
B: moving cholesteryl ester to other lipoproteins is CETP (cholesteryl ester transfer protein).
C: the liver takes up HDL cholesteryl ester through SR-B1 (scavenger receptor B1) and hepatic lipase, not LCAT.
D: chylomicron fat is broken down by LPL (lipoprotein lipase).

Chylomicrons: dietary lipid transport

21. *2026 self-assessment*

A mutation in the apoB-100 gene interferes with the production of an apoprotein with 48% of the amino acid sequence but not the one with 100% of the amino acid sequence. Which of the following lipoproteins would be most affected?

A. IDL

B. VLDL

C. LDL

D. HDL

E. Chylomicrons

Answer: E. Chylomicrons

The protein with 48% of the apoB sequence is **apo B-48**, made only in the intestine and **unique to chylomicrons**. Losing B-48 while B-100 is still made hits **chylomicrons** only.
A, B, C (IDL, VLDL, LDL): intermediate-, very-low- and low-density lipoproteins carry apo B-100 from the liver, which is still made.
D (HDL): high-density lipoprotein; its main protein is apo A-I, and it has no apo B.

22. *2026 Week 13 session*

Sara has been referred to the local lipid clinic for investigation of hypertriglyceridemia associated with defect in the apoproteins that activates Lipoprotein Lipase. Sara is most likely associated with defect in:

A. Apo C II

B. Apo B-48

C. Apo E

D. Apo B100

E. Apo A1

Answer: A. Apo C II

Apo C-II (apolipoprotein C-II) is the activator of **LPL** (lipoprotein lipase) on capillary walls. No C-II → LPL cannot break down triglyceride (TG) in chylomicrons and VLDL (very-low-density lipoprotein) → high TG.
B (Apo B-48): structural protein for building chylomicrons; loss → no chylomicrons, low TG.
C (Apo E): lets the liver take up remnants; it does not activate LPL.
E (Apo A-I): main HDL protein; activates LCAT (lecithin–cholesterol acyltransferase).

23. *2026 self-assessment*

A patient has been diagnosed with an apo-CII deficiency. Which of the following enzymes would be most affected?

A. Lipoprotein lipase

B. HMG-CoA lyase

C. HMG-CoA synthase

D. HMG-CoA reductase

E. 7α-Hydroxylase

Answer: A. Lipoprotein lipase

Apo C-II activates lipoprotein lipase (LPL), the capillary enzyme that breaks down triglyceride in chylomicrons and VLDL (very-low-density lipoprotein). Apo C-II deficiency → LPL inactive → high triglycerides.
B, C (HMG-CoA lyase, synthase): ketone-body and cholesterol-synthesis enzymes (HMG-CoA = 3-hydroxy-3-methylglutaryl-CoA); no apoprotein needed.
D (HMG-CoA reductase): rate-limiting step of cholesterol synthesis; target of statins.
E (7α-hydroxylase): rate-limiting step of bile-acid synthesis.

24. *2026 Canvas practice quiz*

Laboratory results for a patient with uncontrolled type I diabetes mellitus reveals hyperglycemia (560 mg/dL) and hypertriglyceridemia (458 mg/dL). The most likely cause of the increase in triglyceride in this patients may be due to which of the following?

A. Deficiency in apoprotein C-II

B. Increased hepatic triglyceride synthesis

C. Decreased lipoprotein lipase activity

D. Deficiency in LDL receptors

E. Absence of hormone-sensitive lipase

Answer: C. Decreased lipoprotein lipase activity

Insulin is needed to make **lipoprotein lipase (LPL)** in adipose capillaries. In uncontrolled type 1 diabetes (no insulin) **LPL activity falls**, so chylomicron and VLDL (very-low-density lipoprotein) triglyceride is not cleared → hypertriglyceridemia.
A: apo C-II deficiency is a rare inherited defect, not caused by diabetes.
B: the main problem is failed clearance, not overproduction; insulin, which drives liver triglyceride synthesis, is missing.
D: fewer LDL receptors raise LDL cholesterol, not triglyceride.
E: hormone-sensitive lipase is **more** active without insulin.

Cholesterol regulation of LDLR via SREBP2

25. *2026 self-assessment*

How does high [cholesterol] in the liver inhibit uptake of cholesterol by LDL receptors?

- **A.** The SREBP-SCAP complex is proteolytically cleaved from ER/Golgi and inhibits LDLR synthesis
- **B.** SREBP2 cannot move into nucleus and act as a transcription factor to activate LDLR synthesis
- **C.** LDLR is decreased due to PCSK9 degradation of receptors
- **D.** Cholesterol activates SREBP2 to inhibit LDLR synthesis
- **E.** HMG-CoA reductase synthesis is increased leading to a decrease in LDLR synthesis

Answer: B. SREBP2 cannot move into nucleus and act as a transcription factor to activate LDLR synthesis

High cholesterol keeps **SREBP2** (sterol regulatory element-binding protein 2) held in the ER (endoplasmic reticulum) with SCAP (SREBP cleavage-activating protein). It is not cut in the Golgi, so it **cannot enter the nucleus** to switch on LDL receptor genes → fewer receptors.
A: cleavage happens when cholesterol is **low**, and it turns LDL receptor synthesis on.
C: PCSK9 destroys receptors, but it is not how high cholesterol shuts off receptor synthesis.
D: cholesterol blocks SREBP2, which is an activator.
E: reductase synthesis falls too.

Statins and anti-PCSK9 raise LDLR

26. *2026 Canvas practice quiz*

A 60-year-old patient is at his physician's office for a routine health maintenance exam. The patient has a past medical history of osteoarthritis in his right knee and acid reflux that is well-controlled with over the counter medication. On a fasting lipid profile, he is found to have high cholesterol. The patient is started on daily atorvastatin to reduce his risk of cardiovascular disease. What is the major apolipoprotein found on the lipoprotein most directly affected by his statin medication?

- **A.** Apolipoprotein A-I
- **B.** Apolipoprotein B-48
- **C.** Apolipoprotein B-I00
- **D.** Apolipoprotein C-II
- **E.** Apolipoprotein E

Answer: C. Apolipoprotein B-I00

Statins inhibit HMG-CoA reductase → more liver LDL receptors → **LDL (low-density lipoprotein) falls**. The major apolipoprotein of LDL is **apo B-100**, the ligand the LDL receptor recognizes.
A (A-I): HDL (high-density lipoprotein) protein; activates LCAT (lecithin–cholesterol acyltransferase).
B (B-48): chylomicron protein, made in the intestine.
D (C-II): activates lipoprotein lipase on chylomicrons and VLDL (very-low-density lipoprotein).
E (apo E): lets the liver take up remnants; not LDL's main protein.

27. *2026 self-assessment*

How can the introduction of evolocumab for treatment of homozygous FH help to lower blood cholesterol in these patients?

- **A.** It decreases the degradation of the LDLR receptors
- **B.** It decreases the LDLR affinity for oxidized LDL
- **C.** It increases the HMG-CoA activity in the hepatocytes
- **D.** It decreases the SR-A activity in the macrophages
- **E.** It increases the PCSK9 activity on the surface of the hepatocytes

Answer: A. It decreases the degradation of the LDLR receptors

Evolocumab is a monoclonal antibody against **PCSK9** (proprotein convertase subtilisin/kexin 9), a protease that sends LDL receptors (LDLR; LDL = low-density lipoprotein) to the lysosome. Blocking it → **less LDLR degradation** → more receptors recycle to the liver surface → more LDL cleared.
B: LDLR does not bind oxidized LDL; macrophage scavenger receptors do.
C: HMG-CoA reductase is the statin target, and statins inhibit it.
D: SR-A (scavenger receptor A) on macrophages is not the drug's target.
E: the drug blocks PCSK9; it does not raise its activity.

28. *2026 Week 13 session*

Joseph is a 15-year-old boy referred to the cardiologist for dyspnea (difficulty breathing) and chest pain after exertion. Joseph has been on atorvastatin and cholestyramine since age 4. His height and weight are in the normal range. On PE, he has xanthomas and xanthelasmas. His total serum cholesterol is 750 mg/dL. Both his parents have a history of high cholesterol and are being treated with atorvastatin. Which of the following might be the most beneficial to this patient to decrease his cholesterol?

A. Higher levels of cholestyramine

B. Reduced levels of atorvastatin

C. Fat-free diet

D. Insulin injections

E. Anti-PCSK9 therapy

Answer: E. Anti-PCSK9 therapy

Joseph has **homozygous familial hypercholesterolemia** (FH): both parents affected, cholesterol 750 mg/dL, xanthomas in childhood, still high on a statin plus resin. **Anti-PCSK9 antibodies** (evolocumab, alirocumab) block PCSK9 (proprotein convertase subtilisin/kexin 9), so LDL receptors (LDL = low-density lipoprotein) are **recycled instead of degraded** → more LDL cleared.
A (more cholestyramine): a bile-acid resin he already takes; more adds little.
B (less atorvastatin): would raise LDL.
C (fat-free diet): his LDL is genetic and mostly made by the liver; diet barely helps.

LDL oxidation and atherosclerosis

29. *2026 self-assessment · Faculty asked it 4× in 2026*

The uptake of large amounts of cholesterol by macrophages especially in smooth muscle intima eventually results in “foam” cells leading to atherosclerosis. How do the SR-A receptors on these macrophages differ from the LDL receptors (LDLR) on hepatocytes?

A. The SR-A receptors on macrophages are down regulated by free fatty acids

B. Hepatic LDLR synthesis increases under high [cholesterol] via SREBP2

C. The SR-A receptors are not downregulated by high [cholesterol]

D. SR-A are scavenger type receptors and take up excess normal LDL

E. The SR-A receptors on macrophages are down regulated by SREBP2

Answer: C. The SR-A receptors are not downregulated by high [cholesterol]

The **LDL receptor** (LDLR; LDL = low-density lipoprotein) is turned down when cell cholesterol is high (SREBP2, sterol regulatory element-binding protein 2, stays off → no LDLR transcription). The macrophage **SR-A** (scavenger receptor class A) is **not regulated** by cholesterol, so it keeps taking up oxidized LDL → foam cell.
A, E: SR-A is not down-regulated by fatty acids or by SREBP2.
B: backwards; high cholesterol lowers LDLR synthesis.
D: SR-A takes up oxidized (modified) LDL, not normal LDL.

Familial hypercholesterolemia type IIA

30. *2026 Week 13 session*

A 45-year-old man is brought to the emergency department 30 minutes after the sudden onset of crushing chest pain. His father, maternal aunt, and paternal uncle all died of myocardial infarctions under the age of 50 years. Physical examination shows tendinous xanthomas on the hands and thickened Achilles tendons. Serum lipid studies show a total cholesterol concentration of 410 mg/dL, HDL-cholesterol concentration of 30 mg/dL, and triglyceride concentration of 140 mg/dL. The diagnosis of myocardial infarction is made. This patient most likely has a deficiency of which of the following?

A. Apo B48

B. Apo C II

C. HMG-CoA reductase activity

D. LDL receptor

E. Lipoprotein lipase activity

Answer: D. LDL receptor

Total cholesterol 410 mg/dL with normal triglycerides, **tendon xanthomas**, and many relatives with myocardial infarction before 50 = **familial hypercholesterolemia** (FH, type IIa). It is autosomal dominant; 85–90% of cases are a defective **LDL receptor** (LDL = low-density lipoprotein), so LDL cannot be cleared.
A (apo B-48): loss blocks chylomicron formation → low lipids.
B (apo C-II), E (lipoprotein lipase): type I → very high triglycerides (chylomicrons).
C: less HMG-CoA reductase activity would lower cholesterol.

31. *2026 self-assessment*

A 40-year old man presents to a physician for an employment physical. Ocular examination reveals small opaque rings on the lower edge of the iris. Nodular lesions are found on his Achilles tendon. Therapy should be aimed to increase which of the following gene products in hepatocytes?

A. HMG-CoA reductase

B. Apo B-100/E receptor

C. PCAT

D. Apo C-II receptor

E. Apo B-48

Answer: B. Apo B-100/E receptor

Corneal **arcus** (opaque ring at the iris edge) and **Achilles tendon xanthomas** = familial hypercholesterolemia: a defective LDL (low-density lipoprotein) receptor. Therapy (statins, anti-PCSK9) aims to **increase the apo B-100/E receptor**, which is the LDL receptor, on hepatocytes → more LDL cleared.
A (HMG-CoA reductase): more of it would raise cholesterol; statins inhibit it.
C (PCAT = LCAT): lecithin–cholesterol acyltransferase esterifies cholesterol in HDL (high-density lipoprotein); not the defect.
D (apo C-II receptor): none exists; apo C-II activates lipoprotein lipase.
E (apo B-48): chylomicron protein made in the intestine.

32. *Faculty quiz bank 2020–23*

A 28-year-old man was found to have elevated cholesterol levels of 325 mg/dL on a routine checkup. Although his mother has been healthy, his father died of a heart attack at the age of 42. His father also had greatly elevated cholesterol levels throughout his life. The man's physician placed him on lovastatin, and his cholesterol levels dropped to 170 mg/dL. What is his most likely diagnosis?

A. Abetalipoproteinemia

B. Heterozygous FH

C. Homozygous FH

D. Hypertriglyceridemia

E. Lipoprotein lipase deficiency

Answer: B. Heterozygous FH

One affected parent (**autosomal dominant**), cholesterol in the 300s, a father dead of myocardial infarction at 42, and a good fall on lovastatin = **heterozygous familial hypercholesterolemia** (FH). The one working LDL receptor allele (LDL = low-density lipoprotein) can be up-regulated by a statin.
C (homozygous FH): cholesterol ~600–1000 from childhood, two affected parents, little statin response.
A (abetalipoproteinemia): no apo B lipoproteins → very low cholesterol.
D, E (hypertriglyceridemia, lipoprotein lipase deficiency): high triglycerides, not high LDL cholesterol.

Primer 12: comparing lipoprotein classes

33. *2026 Canvas practice quiz*
A 26-year old woman consumed a meal of prime rib, creamy mashed potatoes, salad, bread and a slice of cheesecake. Two hours later, her serum would most likely be milky white due to high levels of:

A. Chylomicrons
B. Fatty acids associated with albumin
C. HDL
D. LDL
E. VLDL

Answer: A. Chylomicrons

Right after a fatty meal, the intestine sends dietary triglyceride into the blood as **chylomicrons**. These large particles scatter light, so serum looks **milky** (lipemic) for a few hours.
B: albumin-bound fatty acids are dissolved and do not cloud serum.
E (VLDL = very-low-density lipoprotein): liver-made; the main fasting triglyceride carrier.
C, D (HDL, LDL): small, low-triglyceride particles; serum stays clear.

34. *2026 Canvas practice quiz*
The level of triglycerides in the blood is an indicator of cardiovascular risk. For routine physical exams, patients are instructed to fast overnight before giving a blood sample for lipid analysis. Under these conditions, most of the triglyceride will be found in which carrier?

A. VLDL
B. Chylomicrons
C. HDL
D. LDL

Answer: A. VLDL

After an overnight fast, chylomicrons from the last meal have been cleared, so most plasma triglyceride is in **VLDL** (very-low-density lipoprotein), made by the liver.
B (chylomicrons): carry dietary triglyceride after meals; normally gone after a 12-hour fast.
C, D (HDL, LDL): carry mainly cholesterol, little triglyceride.

Hyperlipoproteinemia types I–V and lipid panels

35. *2026 Canvas practice quiz*
Which of the following lipoprotein is elevated in patients with type I hyperlipidemia?

A. VLDL
B. IDL
C. LDL
D. Chylomicrons
E. HDL

Answer: D. Chylomicrons

Type I hyperlipoproteinemia = deficiency of **lipoprotein lipase (LPL)** or its activator **apo C-II**. Triglyceride cannot be removed from chylomicrons → **chylomicrons** pile up (triglyceride often >1000 mg/dL, pancreatitis, eruptive xanthomas).
A (VLDL): very-low-density lipoprotein rises in type IV (and in V, with chylomicrons).
B (IDL): intermediate-density lipoprotein rises in type III (apo E2).
C (LDL): low-density lipoprotein rises in type IIa (LDL receptor defect).

Lecture 30 · Eicosanoids: medical aspects (+ Primer 13)

Eicosanoid overview and phospholipid reservoirs

36. *2026 lecture slides*
A cell is stimulated by an inflammatory cytokine, leading to activation of phospholipase A_2 (PLA_2). Which immediate event provides the substrate required for eicosanoid synthesis?

A. Conversion of PGH_2 to prostaglandins

B. Release of arachidonic acid from membrane phospholipids

C. Activation of cyclooxygenase-2

D. Formation of leukotriene A_4

Answer: B. Release of arachidonic acid from membrane phospholipids

Activated **phospholipase A_2 (PLA_2)** hydrolyzes the **sn-2** ester bond of membrane phospholipids and **releases arachidonic acid**, the substrate every eicosanoid pathway needs. Eicosanoids are made on demand from it.
A (PGH_2 → prostaglandins): a later step, after COX (cyclooxygenase) has already made PGH_2.
C (COX-2 activation): COX uses the free arachidonic acid that PLA_2 provides; it does not supply substrate.
D (LTA_4 formation): a downstream 5-lipoxygenase step.

COX pathway and COX-1 vs COX-2

37. *2026 lecture slides*
A researcher inhibits cyclooxygenase activity in a cultured cell. Which molecule can no longer be synthesized directly from arachidonic acid?

A. Leukotriene B_4

B. Lipoxin A_4

C. Prostaglandin H_2

D. Leukotriene A_4

Answer: C. Prostaglandin H_2

Cyclooxygenase (COX) does two steps: arachidonic acid → PGG_2 → **PGH_2** (prostaglandin H_2), the common precursor of prostaglandins, prostacyclin and thromboxanes. Block COX and PGH_2 cannot form.
A (LTB_4), D (LTA_4): made by the 5-lipoxygenase pathway (arachidonic acid → 5-HPETE → LTA_4 → LTB_4); COX block does not stop them.
B (lipoxin A_4): made by lipoxygenases (15-LOX/5-LOX), not COX.

38. *2026 self-assessment*
Amongst the drugs, which interfere with arachidonic acid and eicosanoid metabolism, cyclooxygenase 2 (COX 2) inhibitors:

A. inhibit the constitutive isoform of the enzyme

B. include acetylsalicylic acid (aspirin)

C. are bile acid sequestrants

D. are used to treat type 2 diabetes mellitus

E. prevent prostaglandin release by pro-inflammatory cytokines

Answer: E. prevent prostaglandin release by pro-inflammatory cytokines

COX-2 (cyclooxygenase-2) is the **inducible** isoform. Pro-inflammatory cytokines (IL-1, interleukin-1; TNF-α, tumor necrosis factor-α) switch on its gene in inflamed tissue, and it makes the prostaglandins (PGE_2) of pain and swelling. A COX-2 inhibitor blocks this cytokine-driven prostaglandin output.
A: the mainly **constitutive** isoform is COX-1 (stomach lining, platelets).
B: aspirin is a **non-selective, irreversible** COX-1/COX-2 inhibitor; the selective COX-2 inhibitor is celecoxib.
C, D: bile-acid sequestrants and diabetes drugs are unrelated classes.

LOX pathway and leukotrienes

39. *2026 Week 13 session*
A 25-year-old recent immigrant is suffering from severe pollen allergy. The eicosanoid causing her respiratory distress is most likely to be which of the following?

A. DPPC
B. PGE2
C. TxA2
D. LTC4
E. PGI2

Answer: D. LTC4

In allergic asthma, IgE-activated **mast cells** (and eosinophils) make **LTC_4** (leukotriene C_4), a **cysteinyl leukotriene**. $LTC_4 \rightarrow LTD_4 \rightarrow LTE_4$ act on $CysLT_1$ (cysteinyl leukotriene type 1) receptors → strong **bronchoconstriction**, mucus and airway edema.
A (DPPC, dipalmitoylphosphatidylcholine): lung surfactant; a phospholipid, not an eicosanoid.
B (PGE_2): a bronchodilator that also brakes leukotriene synthesis.
C (TXA_2, thromboxane A_2): platelet aggregation, vasoconstriction.
E (PGI_2, prostacyclin): vasodilation, anti-platelet.

Regulation: cell type and pathway competition

40. *2026 lecture slides*
A patient with asthma develops acute bronchospasm after taking aspirin. Which change in eicosanoid metabolism most directly contributes to this response?

A. Increased prostacyclin synthesis
B. Increased thromboxane synthesis
C. Increased cysteinyl leukotriene synthesis
D. Increased lipoxin synthesis

Answer: C. Increased cysteinyl leukotriene synthesis

Aspirin blocks COX (cyclooxygenase), so more arachidonic acid flows into the **5-lipoxygenase** pathway, and the drop in PGE_2 removes its normal brake on leukotriene synthesis. Result: ↑ **cysteinyl leukotrienes** (LTC_4, LTD_4, LTE_4) → bronchospasm (aspirin-exacerbated respiratory disease).
A (prostacyclin), B (thromboxane): COX products, so aspirin ↓ them.
D (lipoxins): pro-resolving mediators; they calm inflammation and do not cause bronchospasm.

PGI2-TXA2 balance and low-dose aspirin

41. *2026 lecture slides*

Low-dose aspirin reduces the risk of arterial thrombosis primarily by decreasing synthesis of:

A. PGI_2 in endothelial cells

B. PGE_2 in inflammatory tissues

C. TXA_2 in platelets

D. LTC_4 in leukocytes

Answer: C. TXA_2 in platelets

Low-dose aspirin (75–100 mg) **irreversibly blocks platelet COX-1** (cyclooxygenase-1). Platelets lack a nucleus, so **TXA_2** (thromboxane A_2) stays low → ↓ aggregation → ↓ arterial thrombosis.
A (PGI_2): endothelial cells have a nucleus and make new COX, so prostacyclin is relatively spared; lowering it would favor clots.
B (PGE_2): lowering prostaglandin E_2 eases pain and inflammation, not thrombosis.
D (LTC_4): leukotriene C_4 comes from 5-lipoxygenase, which aspirin does not block.

42. *2026 self-assessment*

Patients are advised to stop low dose aspirin regimen at least one week prior to any planned surgery. Which of the following actions of aspirin explains the rationale behind such advice?

A. Aspirin inhibits platelet aggregation mediated by TXA_2

B. Aspirin promotes airway smooth muscle contraction caused by LTC_4

C. Aspirin promotes the anti-inflammatory effect of resolvins

D. Aspirin inhibits the inflammation mediated by PGE_2

E. Aspirin promotes vasodilation caused by PGI_2

Answer: A. Aspirin inhibits platelet aggregation mediated by TXA_2

Low-dose aspirin **irreversibly blocks platelet COX-1** (cyclooxygenase-1), so platelets stop making **TXA_2** (thromboxane A_2), the signal that makes them clump. Platelets have no nucleus and cannot make new COX, so the effect lasts their **7–10-day** life. Stopping a week early lets fresh platelets restore clotting.
E (PGI_2): aspirin does not boost prostacyclin; the endothelium simply keeps making it.
D (PGE_2): a real anti-inflammatory effect, but unrelated to surgical bleeding.
B (LTC_4): aspirin can shunt substrate to leukotrienes (bronchospasm), not a bleeding issue.

Inflammation resolution and SPMs

43. *2026 lecture slides*

Which mediator is classified as a specialized pro-resolving mediator derived from DHA?

A. TXA_2

B. PGE_2

C. Protectin D1

D. LTC_4

Answer: C. Protectin D1

Protectin D1 is a **specialized pro-resolving mediator** (SPM) made from **DHA** (docosahexaenoic acid, omega-3). DHA also gives D-series resolvins and maresins; EPA (eicosapentaenoic acid) gives E-series resolvins; arachidonic acid gives lipoxins.
A (TXA_2): thromboxane A_2; pro-clotting, from arachidonic acid by COX (cyclooxygenase).
B (PGE_2): pro-inflammatory prostaglandin from arachidonic acid.
D (LTC_4): cysteinyl leukotriene; bronchoconstrictor, from arachidonic acid by 5-lipoxygenase.

Drugs targeting eicosanoid pathways

44. *2026 Week 13 session*
A patient is managed for severe asthma with inhaled steroid that blocks the action of an eicosanoid. Steroid blocks the action of an eicosanoid by directly inhibiting enzyme:

A. Cyclooxygenase 1
B. Cyclooxygenases 2
C. Phospholipase A2
D. Lipoxygenase
E. Phospholipase C

Answer: C. Phospholipase A2

Glucocorticoids (steroids) induce **annexin-1** (lipocortin), which **inhibits phospholipase A_2** (PLA_2). PLA_2 frees arachidonic acid from the sn-2 position of membrane phospholipids, so steroids cut substrate for **both** prostaglandins and leukotrienes.
A, B (COX-1, COX-2): cyclooxygenases are blocked by aspirin and other NSAIDs (nonsteroidal anti-inflammatory drugs), downstream of PLA_2.
D (lipoxygenase): blocked by zileuton.
E (phospholipase C): makes second messengers for cell signaling; not the steroid target.

45. *2026 self-assessment*
A 65-year-old female with a 2-year history of arthritis presents with epigastric pain due to a peptic ulcer. She has been self-medicating with aspirin, taking up to 2,000 mg per day for the past six months. Which of the following medications, if taken instead of aspirin, could have minimized her risk of experiencing epigastric pain?

A. Ibuprofen
B. Montelukast
C. Celecoxib
D. Indomethacin
E. Zileuton

Answer: C. Celecoxib

Celecoxib selectively blocks **COX-2** (cyclooxygenase-2, the inflammatory isoform) and spares **COX-1**, which makes the stomach's protective prostaglandins (mucus, bicarbonate, blood flow). So it treats arthritis with **less ulcer risk**.
A (ibuprofen), D (indomethacin): non-selective NSAIDs (nonsteroidal anti-inflammatory drugs) block stomach COX-1 too → ulcers.
B (montelukast), E (zileuton): leukotriene drugs for asthma; they do not treat arthritis pain.

46. *2026 self-assessment*
A 30-year old diabetic patient is in respiratory distress due to seasonal allergies. Since glucocorticoids are contraindicated, which of the following medications is his best choice?

A. Prednisone
B. Alprostadil
C. Zileuton
D. Celecoxib
E. Picotamide

Answer: C. Zileuton

With steroids ruled out, block leukotriene synthesis: **zileuton** inhibits **5-lipoxygenase** → ↓ leukotrienes → ↓ allergic bronchoconstriction.
A (prednisone): a glucocorticoid, excluded by the stem.
B (alprostadil): PGE_1 (prostaglandin E_1); keeps the ductus arteriosus open.
D (celecoxib): COX-2 (cyclooxygenase-2) inhibitor for arthritis.
E (picotamide): thromboxane synthase inhibitor; antiplatelet.

47. *2026 Canvas practice quiz*
A 7-year old boy was hospitalized for an acute asthmatic attack. He was found to have a severe reaction to glucocorticoids and was therefore treated instead with new drugs that specifically block the production or function of leukotrienes. This new group of drugs is most likely to contain compounds that inhibit 5-lipoxygenase and compounds that

A. Inhibit cyclooxygenase 1 (COX1)
B. Inhibit leukotriene receptors
C. Stimulate transcriptional activation by PPAR gamma
D. Stimulate linolenic acid (18 carbon, omega 3 fatty acid) synthesis
E. Inhibit cyclooxygenase 2 (COX2)

Answer: B. Inhibit leukotriene receptors

Two drug groups target leukotrienes: **5-lipoxygenase inhibitors** (zileuton) block their synthesis, and **leukotriene receptor antagonists** (montelukast, zafirlukast, at the CysLT1 receptor) block their action.
A, E (COX-1, COX-2 inhibitors): lower prostaglandins; can shunt arachidonic acid toward more leukotrienes.
C (PPAR-γ): peroxisome proliferator-activated receptor-γ; a diabetes-drug (glitazone) target.
D (linolenic acid synthesis): humans cannot make omega-3 linolenic acid; it is essential (diet).

48. *Heavy 2026 objective*
Which of the following enzymes is correctly matched with its inhibitor?

A. Phospholipase A2 / zileuton

B. 5-lipoxygenase / steroids

C. Thromboxane synthase / picotamide

D. Cyclooxygenase 1 / acetaminophen

Answer: C. Thromboxane synthase / picotamide

Picotamide inhibits **thromboxane synthase** ($PGH_2 \rightarrow TXA_2$, prostaglandin $H_2 \rightarrow$ thromboxane A_2) and blocks the TXA_2 receptor → antiplatelet.
A: phospholipase A_2 is reduced by steroids; zileuton blocks 5-lipoxygenase.
B: pairs swapped: 5-lipoxygenase ← zileuton; steroids act on phospholipase A_2.
D: COX-1 (cyclooxygenase-1) is blocked by aspirin and NSAIDs (nonsteroidal anti-inflammatory drugs); acetaminophen does not meaningfully block it in tissues.

PDA, aspirin uniqueness, clinical consequences

49. *Heavy 2026 objective*
A neonate is born with ductus arteriosus-dependent congenital heart disease. Which of the following prostaglandin analogs would be administered to maintain ductal patency until corrective surgery?

A. Alprostadil

B. Misoprostol

C. Latanoprost

D. Epoprostenol

Answer: A. Alprostadil

Before birth, prostaglandin E keeps the ductus arteriosus (pulmonary artery ↔ aorta) open. **Alprostadil** (**PGE_1**, prostaglandin E_1) keeps it **patent** in ductus-dependent heart defects until surgery.
B (misoprostol): a PGE_1 analog too, but oral, for preventing NSAID (nonsteroidal anti-inflammatory drug) ulcers.
C (latanoprost): $PGF_2\alpha$ analog for glaucoma.
D (epoprostenol): PGI_2 (prostacyclin) for pulmonary hypertension.

50. *2026 Canvas practice quiz*
Which of the following eicosanoids is proinflammatory but also cytoprotective in the GI tract?

A. LTC4

B. PGE2

C. PGI2

D. TXA2

E. HETE

Answer: B. PGE2

PGE_2 (prostaglandin E_2) is **pro-inflammatory** (vasodilation, pain, fever) yet **protects the stomach**: ↓ acid, ↑ mucus and bicarbonate, ↑ mucosal blood flow. Blocking it with NSAIDs (nonsteroidal anti-inflammatory drugs) → ulcers.
A (LTC_4): leukotriene C_4; bronchoconstriction.
C (PGI_2): prostacyclin; mainly vasodilation and anti-platelet.
D (TXA_2): thromboxane A_2; platelet aggregation.
E (HETE): hydroxyeicosatetraenoic acid; attracts neutrophils.